

Getting Serial Number from Zebra RFID API

There are two primary ways to retrieve serial numbers using the Zebra RFID API: during the **Discovery Phase** (from the available reader list) and during the **Connected Phase** (from the reader's radio capabilities).

1. Discovery Phase: `ReaderDevice.serialNumber`

When you are scanning for available readers using `GetAvailableRFIDReaderList()`, you can access the serial number before establishing a full connection. This is useful for identifying hardware (e.g., distinguishing between multiple RFD40 sleds).

```
val readers = Readers(context, ENUM_TRANSPORT.ALL)

try {
    val availableReaders: List<ReaderDevice> = readers.GetAvailableRFIDReaderList()

    availableReaders.forEach { device ->
        // The serial number is a property of the ReaderDevice object
        val deviceSN: String = device.serialNumber
        val deviceName: String = device.name

        Log.d("RFID", "Found Reader: $deviceName | SN: $deviceSN")

        // Example: check for a specific Zebra device
        if (deviceName.startsWith("RFD40") || deviceName.startsWith("TC")) {
            println("Target device found with SN: $deviceSN")
        }
    }
} catch (e: Exception) {
    Log.e("RFID", "Discovery failed", e)
}
```

2. Connected Phase: `ReaderCapabilities.serialNumber`

Once you have selected a `ReaderDevice` and successfully called `connect()`, you can retrieve the internal radio serial number. This is the authoritative serial number from the RFID module.

```
// Connection should be performed in a background thread
CoroutineScope(Dispatchers.IO).launch {
    try {
        val device: ReaderDevice = availableReaders[0]
        val reader: RFIDReader = device.rfidReader
    }
}
```

```

    if (!reader.isConnected) {
        reader.connect()

        // Accessing the serial number via ReaderCapabilities
        // Note: This requires the reader to be CONNECTED
        val radioSN = reader.ReaderCapabilities.serialNumber

        Log.d("RFID", "Connected Reader Radio SN: $radioSN")
        Log.d("RFID", "Host Name: ${reader.hostName}")
        Log.d("RFID", "Model Name: ${reader.ReaderCapabilities.modelName}")
    }
} catch (e: InvalidResponseException) {
    Log.e("RFID", "Connection failed: Invalid response", e)
} catch (e: OperationFailureException) {
    Log.e("RFID", "Connection failed: ${e.results}", e)
}
}

```

3. Alternative: `RFIDReader.serialNumber` (Version Dependent)

In some versions of the Zebra RFID SDK, the `RFIDReader` object itself may expose a `serialNumber` property as a convenience wrapper for the capability. However, using `ReaderCapabilities` is the most reliable cross-version method.

```

// If available in your SDK version:
val sn = reader.serialNumber

```

Summary Comparison

Access Point	Object	Property	State Required
Discovery	<code>ReaderDevice</code>	<code>.serialNumber</code>	No connection needed
Connected	<code>RFIDReader</code>	<code>.ReaderCapabilities.serialNumber</code>	Must be Connected

Best Practices

1. **List Selection:** Use `ReaderDevice.serialNumber` to show the user which device they are about to connect to.
2. **Identification:** Use `ReaderCapabilities.serialNumber` for logging or back-end registration after the connection is solid.

3. **Safety:** Always check `reader.isConnected` before accessing `ReaderCapabilities` to avoid `InvalidResponseException`.